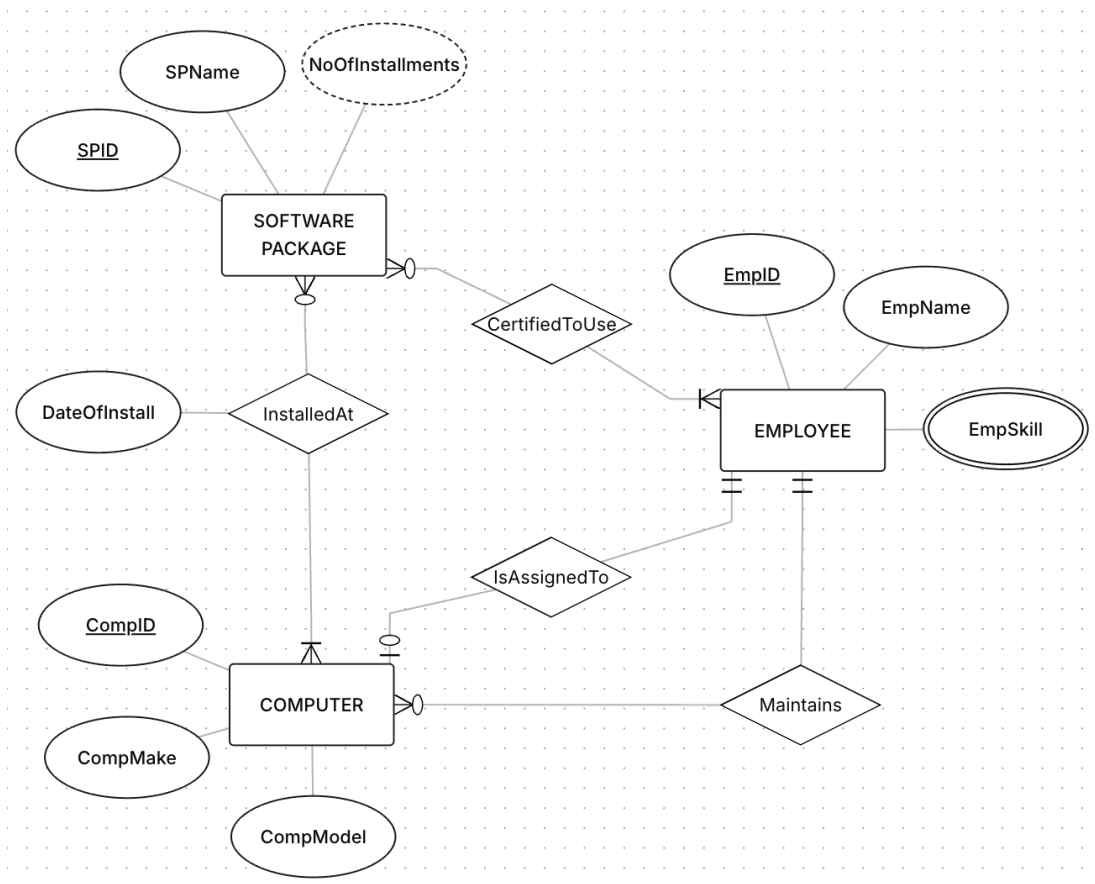


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CS350
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HW 1 ERD

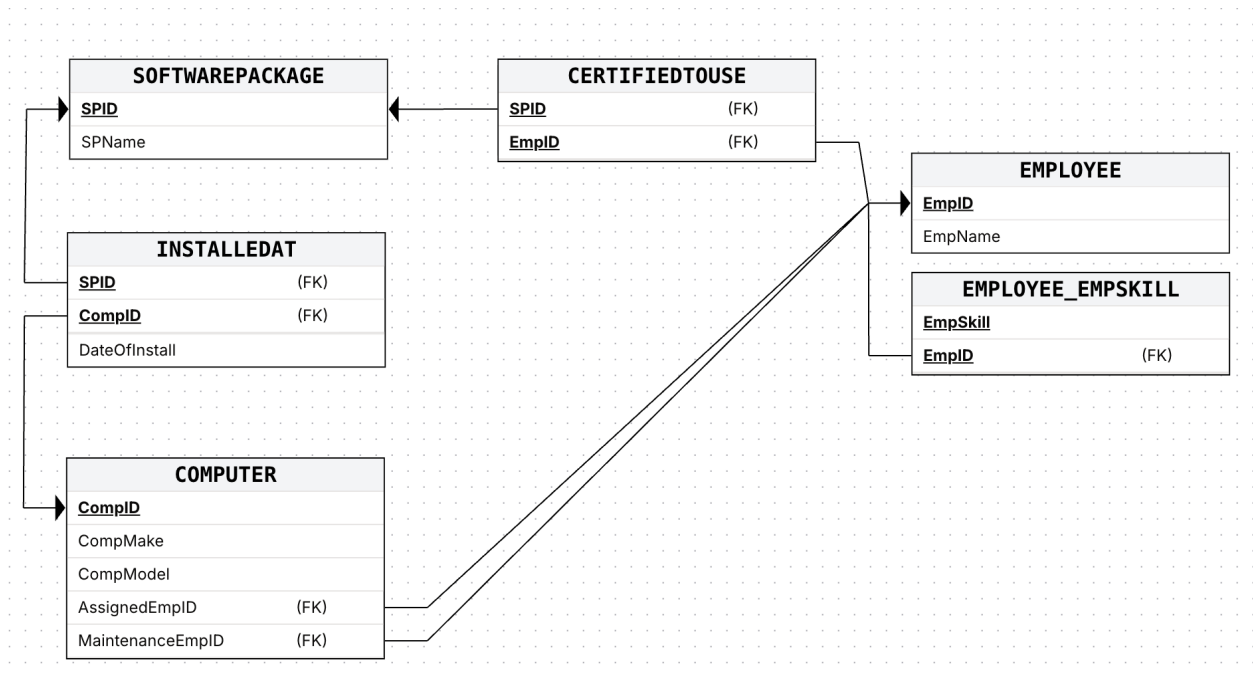
ExoProtect 5 Requirements:

- Each computer can have multiple software packages installed on it, and each software package can be installed on multiple computers.
- An employee can be certified to use several software packages, and a software package can have several employees certified to use it.
- Each employee is assigned exactly one computer, and each computer is assigned to at most one employee.
- An employee may be responsible for maintaining multiple computers, but each computer is maintained by only one employee.
- Employees may possess multiple skills, and this list of skills must be tracked for each employee.

ExoProtect ERD:



ExoProtect Relational Schema:



ExoProtect SQL:

```
CREATE TABLE SOFTWAREPACKAGE
```

```
(  
    SPName CHAR(20) NOT NULL,  
    SPID INT NOT NULL,  
    PRIMARY KEY (SPID)  
);
```

```
CREATE TABLE EMPLOYEE
```

```
(  
    EmpID INT NOT NULL,  
    EmpName CHAR(20) NOT NULL,  
    PRIMARY KEY (EmpID)  
);
```

```
CREATE TABLE COMPUTER
```

```
(
```

```
    CompID INT NOT NULL,
```

```
    CompMake CHAR(20) NOT NULL,
```

```
    CompModel CHAR(20) NOT NULL,
```

```
    AssignedEmpID INT NOT NULL,
```

```
    MaintenanceEmpID INT NOT NULL,
```

```
    PRIMARY KEY (CompID),
```

```
    FOREIGN KEY (AssignedEmpID) REFERENCES EMPLOYEE(EmpID),
```

```
    FOREIGN KEY (MaintenanceEmpID) REFERENCES EMPLOYEE(EmpID)
```

```
);
```

```
CREATE TABLE INSTALLEDAT
```

```
(
```

```
    DateOfInstall DATE NOT NULL,
```

```
    SPID INT NOT NULL,
```

```
    CompID INT NOT NULL,
```

```
    PRIMARY KEY (SPID, CompID),
```

```
    FOREIGN KEY (SPID) REFERENCES SOFTWAREPACKAGE(SPID),
```

```
    FOREIGN KEY (CompID) REFERENCES COMPUTER(CompID)
```

```
);
```

```
CREATE TABLE CERTIFIEDTOUSE
```

```
(
```

```
    SPID INT NOT NULL,
```

```
    EmpID INT NOT NULL,  
    PRIMARY KEY (SPID, EmpID),  
    FOREIGN KEY (SPID) REFERENCES SOFTWAREPACKAGE(SPID),  
    FOREIGN KEY (EmpID) REFERENCES EMPLOYEE(EmpID)  
);
```

```
CREATE TABLE EMPLOYEE_EMPSKILL  
(  
    EmpSkill INT NOT NULL,  
    EmpID INT NOT NULL,  
    PRIMARY KEY (EmpSkill, EmpID),  
    FOREIGN KEY (EmpID) REFERENCES EMPLOYEE(EmpID)  
);
```